

Andrew Pols

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Education

University of Toronto

Sep. 2024 - Present

- HBsc., **Computer Science** | GPA: 4.00 / 4.00
- Relevant coursework: Software Design; Computer Organization; Software Tools and Systems Programming

Experience

Software Developer Intern

May 2026 - May 2027

IBM Canada Ltd.

- Developed a dependency graph visualization for IBM Planning Analytics Workspace, helping clients comprehend their financial planning models by tracking **thousands of objects** while rendering only the relevant subset.
- Engineered the dependency analysis pipeline using TM1 REST APIs, object relationship traversal, and parsing scripts while lazy-loading data, cutting a process that could take **hours to manually trace down to minutes**.
- Built the production front end in React, TypeScript, and React Flow, covering graph layout, object-specific interactions, and filtering for large, complex dependency structures dynamically loaded for each client.
- Worked directly with design, product management, QA, and engineering through an Agile process to shape requirements, resolve technical constraints, and ship the feature to clients.

Projects

ScholarSearch | [Source Code](#)

ReactJS | Python | PostgreSQL | FastAPI

- Built a research paper search engine in a 3-person team, pairing semantic search algorithms with D3.js citation graphs so researchers can see a paper's impact and related work without manually searching pages of references.
- Indexed over **1,000,000** papers using PostgreSQL for storage and ChromaDB for vector embeddings, enabling scalable semantic search with achieved response times of under **200ms** with Redis caching.
- Containerized Postgres, ChromaDB, and Redis with Docker for a consistent development workflow across the team.

FindMySound | [Source Code](#)

ReactJS | Python | Django | Scikit-Learn

- Built a web app that recommends music based on a user's Spotify library, using Spotify's Web API with OAuth 2.0.
- Applied a K-Nearest-Neighbours model with Scikit-Learn to drive recommendations, reaching 95% user satisfaction.
- Handled user accounts through a custom Django user model with JWT-based authentication for secure sessions.
- Exposed RESTful endpoints for authentication, account syncing, and recommendations to power frontend features.

GroupFlow | [Source Code](#)

Java | MongoDB | Swing

- Built a Java desktop app giving university students a central place to schedule meetings and assign tasks.
- Used MongoDB to persist shared data of users, tasks, and meetings, with Java Swing for the desktop interface.
- Followed Clean Architecture and SOLID principles by separating each use case into its own isolated package.
- Collaborated in a team of 6, enforcing CheckStyle and requiring thorough PR reviews before merging code to main.

Technical Skills

- **Programming Languages:** Java, Python, C, JavaScript, TypeScript, SQL, HTML5, CSS
- **Frameworks & Libraries:** React, Node.js, Next.js, Django REST, Flask, React Flow, Scikit-learn, OpenCV
- **Developer Tools:** Git, Jira, Docker, AWS, Azure, Redis, PostgreSQL, MongoDB, Nginx